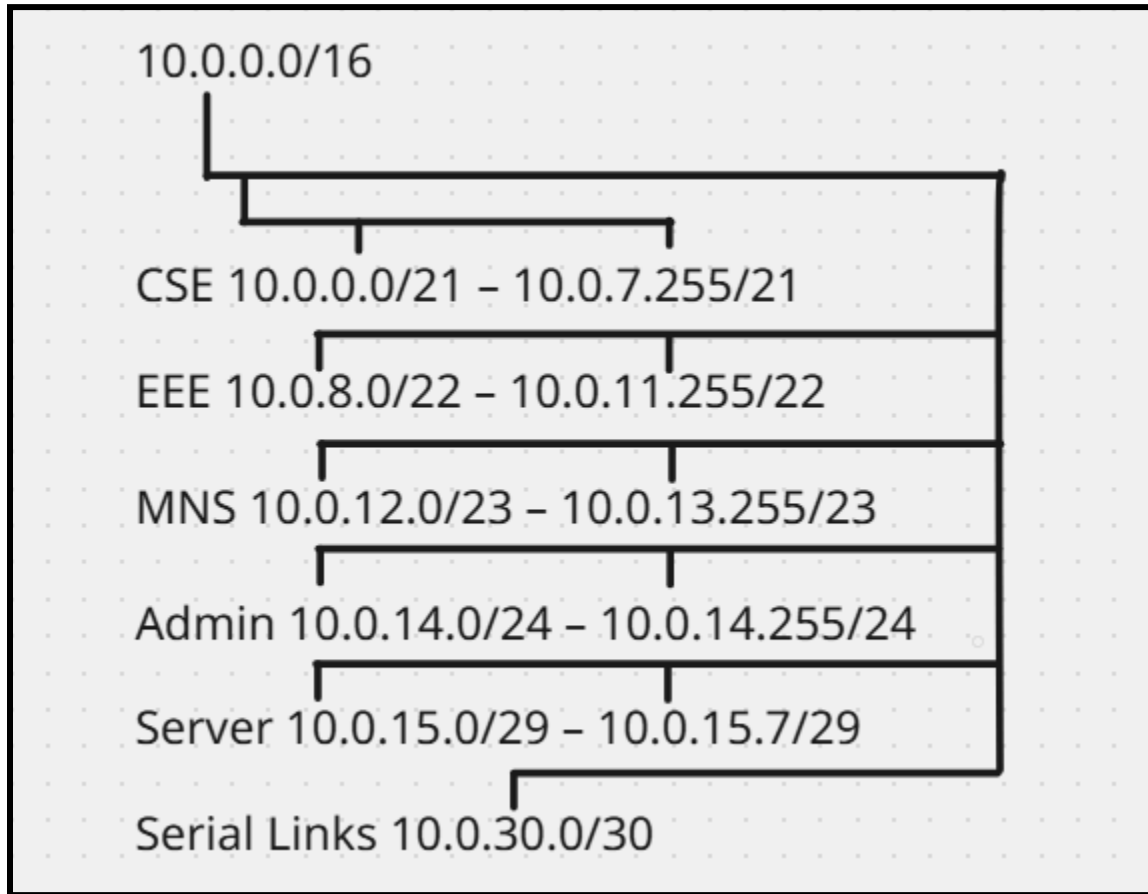


VLSM Tree:



IP Address Table:

Device Name	Interface	IPv4 Address	Subnet Mask	Default Gateway
R-CSE	G0/0 (LAN)	10.0.0.1	255.255.248.0	N/A
	S0/0/0 (to Admin)	10.0.30.1	255.255.255.252	N/A

	S0/0/1 (to Backup)	10.0.30.21	255.255.255.252	N/A
R-EEE	G0/0 (LAN)	10.0.8.1	255.255.252.0	N/A
	S0/0/0 (to Admin)	10.0.30.5	255.255.255.252	N/A
R-MNS	G0/0 (LAN)	10.0.12.1	255.255.254.0	N/A
	S0/0/0 (to Admin)	10.0.30.9	255.255.255.252	N/A
R-Admin	G0/0 (LAN)	10.0.14.1	255.255.255.0	N/A
	S0/0/0 (to CSE)	10.0.30.2	255.255.255.252	N/A
	S0/0/1 (to EEE)	10.0.30.6	255.255.255.252	N/A
	S0/1/0 (to MNS)	10.0.30.10	255.255.255.252	N/A

	S0/1/1 (to Server)	10.0.30.13	255.255.255.252	N/A
R-Server	G0/0 (LAN)	10.0.15.1	255.255.255.248	N/A
	S0/0/0 (to Admin)	10.0.30.14	255.255.255.252	N/A
	S0/0/1 (to Backup)	10.0.30.26	255.255.255.252	N/A
R-Backup	S0/0/0 (to CSE)	10.0.30.22	255.255.255.252	N/A
	S0/0/1 (to Server)	10.0.30.25	255.255.255.252	N/A
Web Server	NIC	10.0.15.2	255.255.255.248	10.0.15.1
DNS Server	NIC	10.0.15.3	255.255.255.248	10.0.15.1
Email Server	NIC	10.0.15.4	255.255.255.248	10.0.15.1

DHCP Server	NIC	10.0.15.5	255.255.255.248	10.0.15.1
Super PC (CSE)	NIC	10.0.7.254	255.255.248.0	10.0.0.1
Super PC (EEE)	NIC	10.0.11.254	255.255.252.0	10.0.8.1
Super PC (MNS)	NIC	10.0.13.254	255.255.254.0	10.0.12.1
Super PC (Admin)	NIC	10.0.14.254	255.255.255.0	10.0.14.1
Super PC (Server)	NIC	10.0.15.6	255.255.255.248	10.0.15.1
Admin PC 1	NIC	10.0.14.2	255.255.255.0	10.0.14.1
Admin PC 2	NIC	10.0.14.3	255.255.255.0	10.0.14.1
CSE PC 1	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>

CSE PC 2	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>
EEE PC 1	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>
EEE PC 2	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>
MNS PC 1	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>
MNS PC 2	NIC	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>	<i>DHCP Assigned</i>

Configuration Commands:

Router Configurations -

Router basic config- main LAN network is 10.0.0.0, WAN is 10.0.30.0/30

#CSE

```

en
erase startup-config
reload
en
conf t
hostname R-CSE
no ip domain-lookup
int g0/0
ip address 10.0.0.1 255.255.248.0
int s0/0/0
ip address 10.0.30.1 255.255.255.252

```

```
interface s0/0/1
ip address 10.0.30.21 255.255.255.252
no shut
copy running-config startup-config
```

#EEE

```
en
erase startup-config
reload
en
conf t
hostname R-EEE
no ip domain-lookup
interface g0/0
ip address 10.0.8.1 255.255.252.0
interface s0/0/0
ip address 10.0.30.5 255.255.255.252
no shut
copy running-config startup-config
```

#MNS

```
en
erase startup-config
reload
en
conf t
hostname R-MNS
no ip domain-lookup
interface g0/0
ip address 10.0.12.1 255.255.254.0
interface s0/0/0
ip address 10.0.30.9 255.255.255.252
no shut
copy running-config startup-config
```

#Admin

```
en
erase startup-config
reload
en
conf t
hostname R-CSE
no ip domain-lookup
interface g0/0
```

```
ip address 10.0.14.1 255.255.255.0
interface s0/0/0
ip address 10.0.30.2 255.255.255.252
clock rate 64000
interface s0/0/1
ip address 10.0.30.6 255.255.255.252
clock rate 64000
interface s0/1/0
ip address 10.0.30.10 255.255.255.252
clock rate 64000
interface s0/1/1
ip address 10.0.30.13 255.255.255.252
clock rate 64000
no shut
copy running-config startup-config
```

#Server

```
en
erase startup-config
reload
en
conf t
hostname R-CSE
no ip domain-lookup
interface g0/0
ip address 10.0.15.1 255.255.255.248
interface s0/0/0
ip address 10.0.30.14 255.255.255.252
interface s0/0/1
ip address 10.0.30.26 255.255.255.252
no shut
copy running-config startup-config
```

#Backup Router

```
en
erase startup-config
reload
en
conf t
hostname R-CSE
no ip domain-lookup
interface s0/0/0
ip address 10.0.30.22 255.255.255.252
clock rate 64000
```

```
interface s0/0/1
ip address 10.0.30.25 255.255.255.252
clock rate 64000
no shut
copy running-config startup-config
```

Configure supercomputer (static)-

IP Address: CSE- 10.0.7.254, EEE- 10.0.11.254, MNS- 10.0.13.254, Admin- 10.0.14.254,
Server-10.0.15.6
Subnet Mask: 255.255.248.0
Default Gateway: 10.0.0.1
DNS Server: 10.0.15.3

#Admin PC config-

IP Address: pc1- 10.0.14.2, pc2-10.0.14.3
Subnet Mask: 255.255.255.0
Default Gateway: 10.0.14.1
DNS Server: 10.0.15.3

Server configuration and pool creation-

#web

IP: 10.0.15.2
Mask: 255.255.255.248
Gateway: 10.0.15.1
HTTP ON
Index.html > Welcome To Brac University

#dns

IP Address: 10.0.15.3
Subnet Mask: 255.255.255.248
Default Gateway: 10.0.15.1
DNS ON
Name: www.bracu.com
Address: 10.0.15.2
Name: bracu.com CNAME

#email

IP Address: 10.0.15.4
Subnet Mask: 255.255.255.248
Default Gateway: 10.0.15.1

#dhcp

IP: 10.0.15.5

Mask: 255.255.255.248

Gateway: 10.0.15.1

DHCP ON

Pool created for CSE, EEE, and MNS

****cse**

Pool Name: CSE_LAN

Default Gateway: 10.0.0.1

DNS Server: 10.0.15.3

Start IP Address: 10.0.0.2

Subnet Mask: 255.255.248.0

Maximum Users: 2040

****eee**

EEE_LAN

10.0.8.1

10.0.15.3

10.0.8.2

255.255.252.0

1020

****mns**

MNS_LAN

10.0.12.1

10.0.15.3

10.0.12.2

255.255.254.0

510

Configure DHCP RELAY-

#CSE

en

confi t

int g0/0

ip helper-address 10.0.15.5

#EEE

en

confi t

int g0/0

```
ip helper-address 10.0.15.5
```

#MNS

```
en  
confi t  
int g0/0  
ip helper-address 10.0.15.5
```

Configure RIP Protocol for CSE, EEE, MNS, Admin-

#CSE

```
ip route 0.0.0.0 0.0.0.0 10.0.30.22 120  
router rip  
version 2  
no auto-summary  
network 10.0.0.0  
network 10.0.30.0  
passive-interface g0/0  
end  
copy run start
```

#EEE

```
router rip  
version 2  
no auto-summary  
network 10.0.8.0  
network 10.0.30.0  
passive-interface g0/0  
end  
copy run start
```

#MNS

```
router rip  
version 2  
no auto-summary  
network 10.0.12.0  
network 10.0.30.0  
passive-interface g0/0  
end  
copy run start
```

#Admin

```
ip route 10.0.15.0 255.255.255.248 10.0.30.14  
router rip
```

```
version 2
no auto-summary
network 10.0.14.0
network 10.0.30.0
passive-interface g0/0
end
copy run start
```

#Server

```
ip route 10.0.0.0 255.255.248.0 10.0.30.13
ip route 10.0.8.0 255.255.252.0 10.0.30.13
ip route 10.0.12.0 255.255.254.0 10.0.30.13
ip route 10.0.14.0 255.255.255.0 10.0.30.13
ip route 0.0.0.0 0.0.0.0 10.0.30.13
ip route 10.0.0.0 255.255.248.0 10.0.30.25 10
```

#Backup

```
ip route 10.0.15.0 255.255.255.248 10.0.30.26
ip route 10.0.0.0 255.255.248.0 10.0.30.21
end
copy run start
```

Configure PCs of CSE, EEE, MNS-

Turning on DHCP

List of any assumptions:

1. CSE, EEE, and MNS buildings can't directly communicate, but can through Admin building.